RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

DEPARTMENT OF PHYSICAL SCIENCES

MAT 3222 – Numerical Analysis II

Tutorial 03

Consider all the values for 5 decimals.

1. Suppose that $y = y(t)$ is the solution to the initial value problem

   $\frac{dy}{dt} = \left(\frac{1}{1+t^2}\right) - 2y,$ $\quad$ y(0)= 0.6

   Carry out two-time steps of Euler's method with a step size of $h = 0.1$ so as to obtain approximations to y (0.1) and y (0.2).

2. Suppose that $y = y(t)$ is the solution to the initial value problem

   $\frac{dy}{dt} = t - 2y,$ $\quad$ y(0)= 1

   Carry out two-time steps of Euler's method with a step size of $h = 0.1$ so as to obtain approximations to y (0.1) and y (0.2).

3. Suppose that $y = y(t)$ is the solution to the initial value problem

   $\frac{dy}{dt} = 1 - y + t$ $\quad$ y(0)= 0.5

   Carry out two-time steps of trapezium method with a step size of $h = 0.2$ so as to obtain approximations to y (0.2) and y(0.4).(without predictor – corrector method)

4. Suppose that $y = y(t)$ is the solution to the initial value problem

$$\frac{dy}{dt} = t^2 - y \qquad y(0) = 1$$

Carry out two-time steps of trapezium method with a step size of $h = 0.2$ so as to obtain approximations to y (0.2) and $y(0.4)$.(without predictor – corrector method)

5. Find the roots of the first characteristic polynomial for each of the linear multi-step methods below and determine the zero – stable, consistent and the order.

   a) $y_{n+2} - y_n = 2hf_{n+1}$

   b) $y_{n+2} + 4y_{n+1} - 5y_n = h(f_{n+2} + 2f_n)$

   c) $y_{n+2} - 2y_{n+1} + y_n = \frac{h}{2}(3f_{n+1} - f_n)$

6. Find the order of the 3-step, explicit linear multi-step scheme:

$$y_{n+3} - y_{n+2} = \frac{h}{12}(23f_{n+2} - 16f_{n+1} + 5f_n)$$

7. Suppose that y = y(t) is the solution to the initial value problem:

$$\frac{dy}{dt} = t^2 - y \qquad y(0) = 1$$

Carry out one time step of the Runge-Kutta method (RK4) with a step size of h = 0.1 so as to obtain an approximation to $y(0.1)$.

8. Suppose that $y = y(t)$ is the solution to the initial value problem

$$\frac{dy}{dt} = t^2 - y \qquad y(0) = 1$$

Carry out two-time steps of trapezium method with a step size of $h = 0.2$ so as to obtain approximations to y (0.2) and $y(0.4)$.(using predictor – corrector method)

<u>Submit the tutorial on or before 09/10/2026</u>